Oliver Johnson

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Education

Case Western Reserve University — *Cleveland, OH*
B.S. Physics, Biophysics Concentration | Expected May 2029 — *August 2025–Present*

- GPA: 4.00/4.00; 2026 SSRP Awardee; Dean's High Honors (2×); Varsity Swimming & Diving

The Hotchkiss School — *Lakeville, CT*
High School Diploma — *August 2021–May 2025*

- High Honor Roll (4×); Dormitory Proctor; Varsity Swimming Captain; Senior Faculty Prize

Research Experience

Wang Research Group, Case Western Reserve University — *Cleveland, OH*
Undergraduate Research Assistant — *October 2025–Present*

- Develop DEMON (Dynamic Exploration of Molecular Simulations Through Optimized Navigation), a reinforcement learning framework that allocates simulation effort across an ensemble of molecular trajectories to explore underrepresented configurations without predefined collective variables.
- Implemented discrete Soft Actor-Critic algorithm and developed training infrastructure including checkpointing, warm-start, replay buffer management, and evaluation pipelines.
- Implemented and benchmarked molecular trajectory encoders combining E(3)-equivariant PaiNN/EGNN spatial representations, rotary temporal Transformers, and permutation-equivariant cross-replica attention.
- Reimplemented the model-system Langevin simulation pipeline in JAX using a JIT-compiled lax.scan integrator and vmap-parallel replica stepping, improving simulation throughput by up to $\approx 190\times$ versus the prior implementation.
- Extended OpenMM parallel-walker execution and implemented graph-aware, memory-mapped replay supporting fixed and variable molecular topologies.
- Diagnosed replica-order dependence by reordering walkers and comparing predicted Q-values, then redesigned the trajectory encoder so permuting walkers produces the corresponding permutation of action values.

Selected Presentations

Johnson, Oliver; Brahmasandra, Arnav; Kosliya Muhammed, Sulekha; Wang, Yihang. "DEMON: Dynamic Exploration of Molecular Simulations Through Optimized Navigation." Poster Presentation, UChicago & Caltech AI+Science Conference, October 2026.

Johnson, Oliver; Brahmasandra, Arnav; Kosliya Muhammed, Sulekha; Wang, Yihang. "DEMON: Dynamic Exploration of Molecular Simulations Through Optimized Navigation." Poster presentation, Case Western Reserve University Intersections, July 2026.

Work Experience

Mailbox Restoration Service — *New Canaan, CT*
Founder & Operator — *Summer 2025*

- Founded and operated an on-demand mailbox-restoration service following identification of the underserved market for express restorations; set pricing models to account for uncertain market size, scheduled client work, and ran targeted digital and print marketing campaigns.

New Canaan Land Trust — *New Canaan, CT*
Summer Stewardship Intern — *Summer 2024*

- Maintained a 400-acre trail network; designed trail lighting for special events and helped revise parking locations to optimize land use.

Honors & Awards

2026 CWRU Sponsored Summer Research Program Awardee | 2025 Senior Faculty Prize, The Hotchkiss School
Five-time Connecticut Swimming Top 16 Athlete | 2025 Connecticut Swimming Scholar Athlete | Three-time YMCA National Championship Qualifier

Technical Skills

ML & RL: PyTorch, Geometric Deep Learning, Off-policy RL
Computing: Linux, Slurm, NumPy, JAX, HPC/GPU workflows
Simulation: OpenMM, MD simulation
Engineering: Performance profiling, Git/GitHub